



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

GECKEL:MUSSEL-GECKO HYBRID ADHESIVE FOR MEDICAL BANDAGES

**BTA0104- Biology and Environmental Science for
Engineers with Sustainable Solutions**

GUIDED BY

Dr. SWATHI

Dr. ALEYAMMA ALEXANDER

DONE BY

Kaviya.G(192524281)

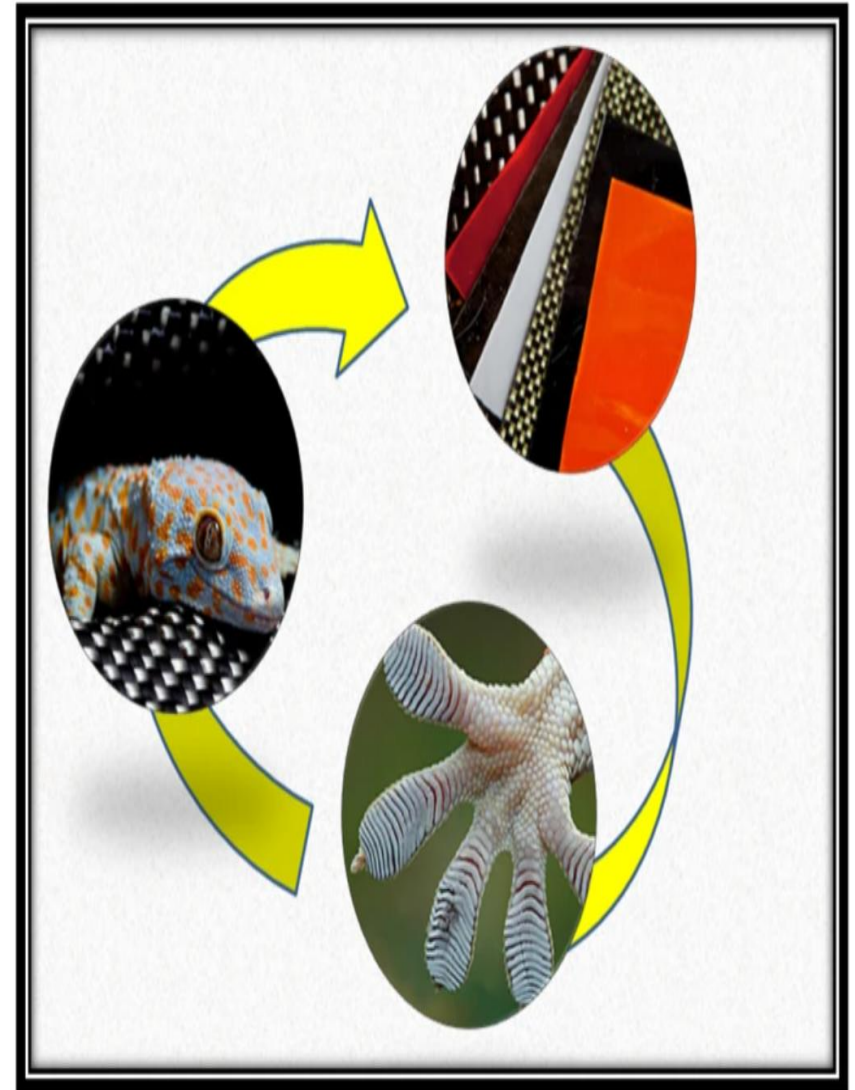
B .Tech (AI & DS)



INTRODUCTION



- Medical bandages are used to cover wounds and help them heal safely.
- Normal bandages do not stick well on wet or sweaty skin and may fall off.
- Removing normal bandages can cause pain and damage to the skin.
- Scientists got inspiration from nature, especially from gecko feet and mussel glue.
- Geckel is a bio-inspired adhesive that combines both to make better medical bandages.

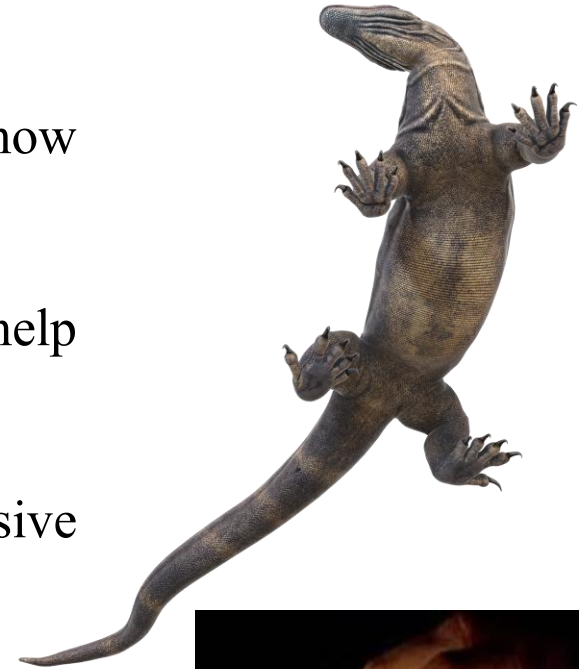




OBJECTIVE



- To study how gecko feet stick to surfaces and how mussels create natural glue.
- To understand how nature-inspired ideas can help solve problems in medical applications.
- To design a strong but skin-friendly adhesive bandage that does not hurt when removed.
- To create a bandage that works well on wet or sweaty skin.
- To improve patient comfort and safety during wound healing.





LITERATURE SURVEY



ENGINEER TO EXCEL

S.No	Author & year	Title of Study	Problem Addressed	Methodology	Key Findings	Limitations
1.	Lee et al., 2016	Gecko–Mussel Inspired Wet Adhesive	Poor adhesion of bandages on wet skin	Gecko microstructure + mussel polydopamine	Strong adhesion on wet tissue	Complex fabrication
2.	Kim & Park, 2017	Mussel-Inspired Medical Adhesives	Skin irritation from synthetic adhesives	Polydopamine-based bio-adhesive coating	Good bio compatibility and wet adhesion	Limited mechanical strength
3.	Chen et al., 2018	Gecko-Inspired Micro-Patterned Adhesives	Bandage failure during skin movement	Micro-patterned elastomer surfaces	Flexible and reusable adhesion	Weak performance in wet conditions
4.	Wang et al., 2019	Hybrid Bio-Adhesives for Medical Use	Single-mechanism adhesives are ineffective	Structural + chemical adhesion system	Strong adhesion on dry and wet skin	Limited long-term testing



PROBLEM IDENTIFICATION AND ANALYSIS



- Normal medical bandages do not stick properly on wet
- Bandages often fall off during movement, causing poor wound protection.
- Removing regular bandages can cause pain and skin irritation.
- Chemical adhesives used in bandages may damage sensitive skin.
- There is a need for a strong, safe, and painless adhesive for medical use.





POSSIBLE SOLUTIONS



➤ **Bio-Inspired Hybrid Adhesive**

Uses gecko-inspired dry adhesion and mussel-inspired wet adhesion.

➤ **Works on Dry & Wet Skin**

Provides strong sticking even on wet, sweaty, or moist skin.

➤ **Strong Yet Painless Removal**

Sticks securely but removes easily without pain or skin damage.

➤ **Skin-Friendly & Safe**

Made from biocompatible, non-toxic materials, suitable for sensitive skin.

➤ **Stable During Body Movement**

Does not fall off during walking, bending, or daily activities



IMPLEMENTATION

1. Bio-Inspired Research

Study gecko-like microstructures & mussel-inspired chemistry.

2. Material Development

Formulate hybrid adhesive inspired by geckos & mussels.

3. Nanostructure Fabrication

Create gecko-like nanostructures on adhesive surface.

4. Testing & Optimization

Test & optimize adhesive strength, flexibility, & safety.

5. Advanced Medical Bandages

Produce advanced medical bandages with enhanced adhesion.

5. Advanced Medical Bandages





ADVANTAGE



ENGINEER TO EXCEL



Painless Removal

Ensuring the **adhesive** can be removed without causing pain or damage to the **skin**.



Effective on Dry & Wet Skin

Designing the **adhesive** to stick well on both dry and wet skin.



Non-Irritating Formula

Ensuring the **adhesive** does not cause skin irritation or allergic reactions.



Strong Yet Removable

Creating a strong adhesive that holds securely yet peels off easily using gecko-like **nanostructures**.



Safe & Affordable for Medical Use

Making the **adhesive** **safe**, biocompatible, and **cost-effective** for widespread medical applications.



Scalable & Consistent Production

Developing a **manufacturing** process that is scalable, consistent, and reliable.

CONCLUSION

- Geckel is a bio-inspired adhesive made using ideas from gecko and mussel.
- It helps medical bandages stick well on both dry and wet skin.
- The adhesive can be removed easily without causing pain or skin damage.
- This project shows how nature can help solve medical problems.
- Geckel adhesive can improve comfort and safety in wound care.





REFERENCES



1. Liu, Y., Wang, H., Li, J., Li, P., & Li, S. (2024). Gecko-Inspired Controllable Adhesive: Structure, Fabrication, and Application. *Biomimetics*, 9(3), 149
2. Guo, Q., Chen, J., Wang, J., Zeng, H., & Yu, J. (2020). Recent progress in synthesis and application of mussel-inspired adhesives. *Nanoscale*, 12, 1307–1324.
3. Nature-inspired adhesive systems (2024). *Chemical Society Reviews* — discusses bioinspired adhesives and their biomedical use.— Good reference for medical adhesive overview and possibilities.
4. Bio-based and bio-inspired adhesives from animals and plants for biomedical applications (2022). *Materials Today Bio* — overview of mussel and animal-inspired adhesives for medicine.
5. Marine-inspired polymers in medical adhesion (2021). *PubMed article* — examines mussel adhesion principles applied to medical adhesives

The background of the image is a dense, repeating pattern of dark blue, almost black, leaves. The leaves are elongated and have a glossy texture, with some showing lighter blue highlights. They are arranged in a way that creates a sense of depth and movement. A semi-transparent dark blue rectangular box is centered over the image, serving as a backdrop for the text.

Thank you